

QUANTUM BASICS • BEGINNER NOTES

Sections 03–04

GUIDED LESSON • LEARN BY EXPERIMENTING

Quantum Circuits

Start with the simplest possible circuit, see what happens when we do nothing, and then use the X gate to deliberately change a qubit from one basis state to the other.

03 Try a Circuit With No Gates

Sometimes the most useful experiment is doing nothing

1. What is a quantum circuit?

A **quantum circuit** is a sequence of operations performed on one or more qubits. We draw a circuit from **left to right**. The horizontal line represents the qubit, while boxes placed on the line represent operations or gates.

Think of the circuit as a timeline

The qubit enters from the left, moves through the circuit, experiences whatever operations we place on it, and then reaches the measurement at the right. If there are no gates between the beginning and the end, the qubit simply travels through unchanged.

2. The simplest possible circuit

Let us begin with a qubit prepared in the state **|0>**. We will not apply any gate to it. There is therefore nothing in the circuit that changes its state.

CIRCUIT 1 • NO GATES

$|0\rangle \quad |0\rangle$

The qubit enters as $|0\rangle$ and leaves as $|0\rangle$.

3. What happened?

Nothing happened to the qubit — and that is exactly the point of the experiment. Since no gate was applied, the state was not changed.

$$|0\rangle \rightarrow |0\rangle$$

No gate means no operation. If a qubit begins in $|0\rangle$ and we apply no operation, it remains in $|0\rangle$ until something else acts on it.

4. What happens when we measure it?

A measurement asks us to observe the qubit in the computational basis. Since the qubit is in the state $|0\rangle$, measuring it gives **0 with 100% probability**.

$$P(0) = 1 \quad P(1) = 0$$

WORKED EXAMPLE

Start with: $|0\rangle$

Apply no gates: $|0\rangle \rightarrow |0\rangle$

Measure: **Result = 0**

Probability: **100% for 0** and **0% for 1**.

5. Now try starting with $|1\rangle$

The same idea works if the qubit starts in $|1\rangle$. Again, we do not apply any gates. Therefore, the state stays exactly as it is.

CIRCUIT 2 - NO GATES STARTING FROM $|1\rangle$ $|1\rangle$ $|1\rangle$ *With no gate, $|1\rangle$ remains $|1\rangle$.*

$$|1\rangle \rightarrow |1\rangle$$

6. Measurement probabilities

Since the qubit is definitely in $|1\rangle$, measuring it gives the result **1 with 100% probability**. The probability of obtaining 0 is zero.

COMPARE THE TWO CASES

Start with $|0\rangle$: **100% $\rightarrow$ 0**

Start with $|1\rangle$: **100% $\rightarrow$ 1**

In both cases, the absence of gates means that the state is unchanged.

7. Why try a circuit with no gates?

It gives us a **baseline**. Before learning what a gate does, we should know what happens when no gate does anything at all. This makes it much easier to understand the effect of the next circuit we build.

No operation

The qubit continues in its current state. Nothing deliberately changes its amplitudes.

Measurement

At the end, measurement tells us whether we obtained 0 or 1.

8. The circuit is not the same as the measurement

It is important to separate two ideas: **operations** and **measurement**. A gate changes a quantum state. Measurement reads an outcome from the state. In our no-gate circuit, there is no operation between preparation and measurement.

- 1 Prepare the qubit in $|0\rangle$ or $|1\rangle$.
- 2 Apply no gates.
- 3 The qubit remains in its original state.
- 4 Measure the qubit.

9. What if the qubit starts in superposition?

The same principle applies. If we somehow prepare a qubit in the state $(|0\rangle + |1\rangle)/\sqrt{2}$ and then apply no gates, the state remains the same.

CIRCUIT 3 - SUPERPOSITION WITH NO GATES

$|+\rangle$

$|+\rangle$

No gate means the $|+\rangle$ state is left unchanged.

Because the state is still $(|0\rangle + |1\rangle)/\sqrt{2}$, measurement still gives a **50% chance of 0** and a **50% chance of 1**.

Starting state	No gate	Measurement
$ 0\rangle$	$ 0\rangle$	0 with 100%
$ 1\rangle$	$ 1\rangle$	1 with 100%
$ +\rangle$	$ +\rangle$	0 or 1 with 50% each

KEY IDEA

A circuit does not automatically change a qubit. **If there is no gate, there is no deliberate state-changing operation.** The qubit simply continues in the state in which it was prepared until measurement or another operation occurs.

QUICK CHECK

1. What does the horizontal line in a circuit represent?

→ The qubit as it moves through the circuit.

2. What happens if there are no gates?

→ The state is not deliberately changed.

3. If $|0\rangle$ enters a no-gate circuit, what leaves?

→ $|0\rangle$.

4. What is the probability of measuring 0 from $|0\rangle$?

→ 100%.

04 Prepare One With X

Use the X gate to flip a qubit

1. Meet the X gate

The **X gate** is one of the simplest quantum gates. It changes $|0\rangle$ into $|1\rangle$ and $|1\rangle$ into $|0\rangle$. Because of this behaviour, it is often compared to a classical **NOT gate**.

The basic rule

Think of X as a **quantum flip**. If the qubit is in $|0\rangle$, X flips it to $|1\rangle$. If the qubit is in $|1\rangle$, X flips it to $|0\rangle$.

2. X acting on $|0\rangle$

Start with a qubit in $|0\rangle$. Apply the X gate. The output becomes $|1\rangle$.

CIRCUIT 4 - X FLIPS $|0\rangle$ $|0\rangle$  $|1\rangle$ *Applying X to $|0\rangle$ produces $|1\rangle$.*

$$X|0\rangle = |1\rangle$$

3. What happens when we measure?

After X has changed $|0\rangle$ into $|1\rangle$, the qubit is in the definite state $|1\rangle$. Therefore, measuring it gives **1 with 100% probability**.

WORKED EXAMPLEInitial state: $|0\rangle$ Apply X: $X|0\rangle = |1\rangle$ Measure: **1**Probability: **100% for 1**.**4. X acting on $|1\rangle$**

X works in the opposite direction too. If the qubit begins in $|1\rangle$, applying X changes it into $|0\rangle$.

CIRCUIT 5 - X FLIPS $|1\rangle$ $|1\rangle$  $|0\rangle$ *Applying X to $|1\rangle$ produces $|0\rangle$.*

$$X|1\rangle = |0\rangle$$

5. X behaves like a NOT operation

In classical computing, a NOT operation changes $0 \rightarrow 1$ and $1 \rightarrow 0$. The X gate performs exactly this flip on the two computational basis states.

Input	X gate	Output
0	Flip	1
1	Flip	0

6. The matrix behind X

Remember that a quantum gate is a **matrix**. The X gate is represented by the following matrix:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

This matrix tells us how the amplitudes of a qubit are transformed when the X gate is applied. We can verify its behaviour by multiplying it by the basis-state vectors.

7. Writing $|0\rangle$ and $|1\rangle$ as vectors

In matrix form, the two computational basis states can be written as column vectors.

$|0\rangle$

The vector representation is:

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}^T$

$|1\rangle$

The vector representation is:

$[0, 1]^T$

8. Verify $X|0\rangle = |1\rangle$

We can use ordinary matrix multiplication to check what the X gate does to $|0\rangle$.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

So the matrix calculation agrees with the circuit rule: **$X|0\rangle = |1\rangle$** .

9. Verify $X|1\rangle = |0\rangle$

Now apply the same matrix to $|1\rangle$.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

10. Applying X twice

What happens if we apply the X gate twice? The first X flips the qubit. The second X flips it back.

CIRCUIT 6 - TWO X GATES

The first X gives $|1\rangle$; the second X returns the qubit to $|0\rangle$.

$$X(X|0\rangle) = |0\rangle$$

Important: Applying X twice is equivalent to doing nothing. In matrix language, this means $X^2 = I$, where I is the identity matrix.

11. The identity operation

The **identity** operation leaves a quantum state unchanged. It is represented by the identity matrix I . So a no-gate circuit behaves like applying the identity:

$$I|0\rangle = |0\rangle \quad I|1\rangle = |1\rangle$$

12. X also acts on superpositions

The X gate is not only useful for $|0\rangle$ and $|1\rangle$. It also acts on superpositions by transforming their components.

$$X(\alpha|0\rangle + \beta|1\rangle) = \alpha|1\rangle + \beta|0\rangle$$

Notice that the coefficients remain attached to the corresponding amplitudes, while the basis states are exchanged. This is the more general version of the “flip” idea.

13. Example with a superposition

Consider the state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Applying X gives:

$$\begin{aligned}
 X|+\rangle &= X(|0\rangle + |1\rangle)/\sqrt{2} \\
 &= (|1\rangle + |0\rangle)/\sqrt{2} \\
 &= (|0\rangle + |1\rangle)/\sqrt{2} \\
 &= |+\rangle
 \end{aligned}$$

A surprising result

The $|+\rangle$ state is unchanged by X . This does **not** mean X does nothing. X still swaps the $|0\rangle$ and $|1\rangle$ components; in this particular symmetric state, the result happens to be the same state.

14. A complete beginner circuit

We can now combine everything we have learned. Start with $|0\rangle$, apply X , and then measure.

CIRCUIT 7 - PREPARE $|1\rangle$ FROM $|0\rangle$



X changes the prepared $|0\rangle$ into $|1\rangle$ before measurement.

15. Step through the circuit

- 1 The qubit begins in the state $|0\rangle$.
- 2 The X gate is applied.
- 3 X flips the state: $|0\rangle \rightarrow |1\rangle$.
- 4 The qubit is measured.

5 The result is **1 with 100% probability**.

16. Compare no gate with X

Now the difference between the two circuits becomes clear. A circuit with no gate leaves the state alone. The X gate deliberately flips the computational basis state.

Circuit	Initial state	Final state	Measurement
No gate	$ 0\rangle$	$ 0\rangle$	0 with 100%
X	$ 0\rangle$	$ 1\rangle$	1 with 100%
No gate	$ 1\rangle$	$ 1\rangle$	1 with 100%
X	$ 1\rangle$	$ 0\rangle$	0 with 100%

17. The most important X-gate rules

Rule 1

$X|0\rangle = |1\rangle$

X flips zero to one.

Rule 2

$X|1\rangle = |0\rangle$

X flips one to zero.

Rule 3

$X^2 = I$

Two X gates cancel each other.

Rule 4

X is a quantum gate represented by a matrix. It transforms the qubit state.

KEY IDEA

The X gate is the simplest example of a quantum operation that changes a qubit. It behaves like a NOT operation on the computational basis: $|0\rangle \leftrightarrow |1\rangle$. Starting with $|0\rangle$ and applying X prepares $|1\rangle$, while applying X twice brings the qubit back to its original state.

QUICK CHECK

1. What does the X gate do to $|0\rangle$?

→ $X|0\rangle = |1\rangle$.

2. What does X do to $|1\rangle$?

→ $X|1\rangle = |0\rangle$.

3. What happens if X is applied twice?

→ The original state is restored.

4. What is X^2 ?

→ The identity operation I.

5. If $|0\rangle \rightarrow X \rightarrow$ measurement, what is the result?

→ 1 with 100% probability.

6. Why is a no-gate circuit useful?

→ It gives us a baseline: without an operation, the quantum state is not deliberately changed.